Condition, Don't Remove: Speed-Conditioned Normal Modeling for Unsupervised Fault Detection in Rotating Machinery

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Preprint. Draft generated from the EngineKnock experiment registry.

Abstract

Unsupervised fault detection for rotating machinery trains on healthy data alone, and its central obstacle is operating-condition shift: a healthy machine produces markedly different vibration and acoustic spectra at different rotation speeds, so faults hide inside the speed-induced spread of the normal class. Existing methods either remove speed through order tracking and speed-invariant representations, or treat it as a discrete domain label. We instead condition the normal model on the continuous speed covariate that rotating machinery already measures through its tachometer, realized simply by subtracting a local speed-conditional normal mean, estimated by nearest neighbours in speed, before a standard reconstruction-error detector. On MaFaulDa (accelerometers, microphone, tachometer, five fault types, 12 to 60 Hz), conditioning raises pooled ROC-AUC from 0.956 to 0.989 at matched speed and from 0.897 to 0.937 on held-out speeds, winning on all five fault types in both regimes; the order-tracking invariance baseline never beats the unconditional detector under speed shift. The gain is backbone-agnostic across PCA and autoencoder detectors, holds across accelerometer and microphone channels, turns multi-channel fusion from harmful into beneficial, and collapses when speed labels are permuted (AUC 0.961 to 0.419). The principle transfers to the MIMII-DG acoustic benchmark, including a discrete operating attribute. For speed variation, condition, don't remove.

1 Introduction

Vibration and acoustic signatures are the standard evidence for the mechanical health of rotating machinery. In deployment the operative formulation is unsupervised: a detector is trained only on data from a healthy machine and must flag any deviation, because labeled fault examples are scarce, expensive, and rarely cover the fault that eventually occurs. This normal-only framing is the one adopted by the DCASE unsupervised anomalous-sound-detection challenges and by most machine-condition autoencoder work.

The dominant obstacle in this setting is operating-condition shift. A healthy machine at 1500 rpm and the same machine at 3500 rpm produce spectra whose peaks sit at entirely different frequencies; a normal-only model that pools across speeds therefore learns a wide, diffuse notion of "normal," and a genuine fault at an intermediate speed can score no more anomalous than a healthy sample at an unseen speed. The challenge is recognized: the DCASE 2021 and 2022 tasks explicitly frame speed and load variation as the cause of a *domain shift* between training and test.

Two families of remedies exist. *Invariance* methods remove the operating condition from the representation, most commonly by order tracking: resampling the signal against shaft angle so that rotation-locked components land in fixed order bins regardless of speed. *Discrete-domain* methods keep the operating condition but encode it as a categorical attribute or a source/target domain label, as in the DCASE domain-generalization setup. Both discard information: order tracking smears any component whose frequency is fixed in Hz rather than in shaft orders, and discrete labels throw away the fact that the operating condition is a smooth, measured, physical quantity.

We argue for a third option that is both simpler and stronger: *condition* the normal-only model on the operating covariate itself. Rotating machinery is commonly instrumented with a tachometer, so the rotation speed is available as a continuous measured variable at both training and test time. Modeling the normal manifold *given* the measured speed removes the speed-induced spread that hides faults, without discarding speed-dependent structure and without collapsing the covariate to a handful of discrete bins. Our contributions are:

- A speed-conditional formulation of normal-only anomaly detection that conditions on the continuous tachometer covariate, and a simple realization of it (local speed-conditional mean subtraction followed by a standard reconstruction-error detector).
- Evidence on MaFaulDa that conditioning beats both the unconditional baseline and an order-tracking invariance baseline on all five fault types, in both matched-speed and held-out-speed regimes.
- Ablations showing the effect is backbone-agnostic (PCA and autoencoder) and holds across accelerometer and microphone channels, where conditioning is also what makes multi-channel fusion helpful.
- A covariate-shuffle control establishing that the improvement is causally the speed conditioning, not an artifact of the pipeline.

2 Related work

Unsupervised machine anomaly detection. Normal-only detection of machine faults from sound and vibration is the subject of the DCASE Task 2 series and its datasets MIMII and ToyADMOS. The prevailing recipe is a reconstruction autoencoder or a pretrained-embedding backend with a k-nearest-neighbour or Mahalanobis score. These methods target the normal-vs-anomaly boundary but do not model the operating condition explicitly.

Domain shift and domain generalization. DCASE 2021 introduced anomalous-sound detection under domain shift and named motor speed and signal-to-noise ratio as its causes; DCASE 2022 reframed the task as domain generalization across source and target domains. In these formulations the operating condition enters as a discrete source/target label or as categorical attribute metadata, not as a continuous covariate that the model conditions on.

Speed-invariant fault diagnosis. Order tracking and angular resampling are the classical route to speed robustness and underlie a large supervised bearing-diagnosis literature; recent unsupervised, variable-speed methods pursue the same *invariance* goal with order-tracking autoencoders and order-adaptive subspace learning (OASSL) [9]. Invariance and conditioning are different design choices: invariance removes the covariate, conditioning models the response given the covariate. Our results indicate that for anomaly detection under speed shift, removing speed can be worse than not addressing it at all.

Conditional and contextual anomaly detection. Conditioning a normal model on an auxiliary variable is an established idea: conditional (or contextual) anomaly detection dates to Song et al. [10], and recent detectors condition normalizing flows on spatial position [7] or auto-encoders on a discrete machine identity [8]. What is specific here, and what we have not found addressed, is conditioning a normal-only machine-fault detector on the *measured continuous rotation speed* from a tachometer and evaluating it under operating-speed shift with a direct head-to-head against order-tracking invariance. Our discrete-bin ablation makes the distinction concrete: conditioning on the continuous covariate beats conditioning on a binned (discrete) version of it, which is the form the closest prior work takes.

3 Problem formulation

We are given a training set of healthy windows $\{(x_i, s_i)\}_{i=1}^{n}$, where $x_i \in \mathbb{R}^d is a feature vector and $s_i \in \mathbb{R}$ is the measured operating covariate, here the rotation speed from a tachometer. Training contains no fault examples. At test time, given a window (x, s), a detector outputs an anomaly score $A(x, s)$ that should be larger for faults than for healthy windows. The difficulty is operating-condition shift: the healthy conditional $p(x \mid s)$ varies strongly with s, so a fault at one speed can resemble a healthy window at another. Three model families address this differently. The *pooled* (unconditional) family models $p(x)$ and ignores s. The *invariant* family transforms $x \mapsto \varphi(x)$ so that φ is approximately independent of s, most commonly by order tracking, and models $p(\varphi(x))$. The *conditional* family, which we advocate, models $p(x \mid s)$ directly using the measured covariate. Invariance removes s; conditioning uses it.

4 Method

Features. Each recording carries a once-per-revolution tachometer pulse train. We segment it into non-overlapping windows of R shaft revolutions and take the log-magnitude spectrum of a chosen sensor channel, $x = \log(1 + |\mathcal{F}\{w\}|)$, as the feature. The window speed $s = f_s/\Delta$ is the sampling rate divided by the median tachometer pulse spacing Δ.

Speed-conditional detector. We estimate the local healthy mean at speed s by a k-nearest-neighbour average in the speed variable,

$$m(s) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s)} x_i , \qquad \mathcal{N}_k(s) = \{\text{indices of the } k \text{ training windows with speed nearest } s\},$$

and detect on the residual $r = x - m(s)$. With $\tilde{r}$ the residual standardized by the training-residual mean and standard deviation, and V the top-K principal components of the standardized healthy-training residuals, the anomaly score is the reconstruction error

$$A(x, s) = \| \tilde{r} - VV^{\top}\tilde{r} \|_2^2 .$$

Subtracting $m(s)$ removes the operating-condition component of the healthy spread while preserving fault-induced deviations, and it needs only the tachometer signal that rotating machinery already provides. As $k \to n$ the estimate $m(s)$ tends to the global mean and the detector reduces to the unconditional baseline, a degenerate limit we verify empirically. A discrete-covariate variant replaces the kNN average by the mean of the training bin into which s falls; a stronger backbone replaces the PCA reconstruction error by that of a dense autoencoder trained

on the residuals. The estimator is deliberately simple: the contribution is the conditioning principle, not a particular smoother.

Baselines. The unconditional baseline applies the same reconstruction-error detector to x with no mean subtraction. The invariance baseline applies it to an order-tracked spectrum, obtained by resampling each window from time to shaft angle using the tachometer before the spectrum is taken, so rotation-locked components occupy speed-independent order bins. All three families share the same backbone, window, and feature dimension, so the comparison isolates the treatment of the covariate.

Algorithm 1: speed-conditional detector

1. **Train** (healthy only): compute residuals $r_i = x_i - m(s_i)$; standardize; fit V = top-K PCA of the standardized residuals (or train an autoencoder on them).
2. **Score** (x, s): find $\mathcal{N}_k(s)$ by binary search in the sorted training speeds; form $r = x - m(s)$; standardize; return $A(x, s) = \|\tilde{r} - VV^\top \tilde{r}\|_2^2$.

Training costs one PCA, $O(nd\min(n, d))$; scoring costs a $O(\log n)$ speed lookup plus a $O(dK)$ projection. The method adds only the kNN mean over the unconditional detector, and stores the training windows (or their per-speed summaries) for the lookup.

5 Experimental setup

We use the MaFaulDa machinery-fault database: a rotor test rig instrumented with two tri-axial accelerometers (underhang and overhang bearings), a microphone, and a tachometer, sampled at 50 kHz. It contains a healthy class and five fault classes (horizontal and vertical shaft misalignment, mass imbalance, and underhang and overhang bearing faults), recorded across rotation speeds spanning roughly 12–60 Hz. Unless stated otherwise the sensor channel is the underhang axial accelerometer.

Protocols. Every detector is trained only on healthy windows. We report two regimes. *Matched-speed*: train on a random 60% of healthy windows across all speeds and test the remainder against faults; this is the standard operating-range setting. *Held-out-speed* (leave-one-band-out): partition the speed axis into bands, train the healthy model on all-but-one band, and test healthy and fault windows drawn only from the held-out band; this measures generalization to unseen operating speeds. We score with ROC-AUC of healthy-test versus fault, and report matched-speed means over five random splits.

Controls. Two controls guard the claims. A *normal-versus-normal null* checks that held-out healthy windows score at chance against other held-out healthy windows (AUC ≈ 0.5), ruling out that the detector keys on the split. A *covariate-shuffle* control detrends the healthy-test windows with randomly permuted speed labels; if the gain is genuinely the speed-conditioning, this must collapse the AUC.

6 Results

Table 1. *ROC-AUC per fault type (underhang axial accelerometer). Columns: Base (unconditional), Inv (order-tracking invariance), Cond (ours), for the matched-speed (M) and held-out-speed (H) regimes. M is the mean over five splits; H is the leave-one-band-out mean. Conditioning is best in every cell; the invariance baseline never beats the unconditional baseline under speed shift.*

Fault type	Base (M)	Inv (M)	Cond (M)	Base (H)	Inv (H)	Cond (H)
Horizontal misalignment	0.854	0.873	0.957	0.689	0.627	0.794
Vertical misalignment	0.977	0.953	0.996	0.970	0.872	0.984

Fault type	Base (M)	Inv (M)	Cond (M)	Base (H)	Inv (H)	Cond (H)
Imbalance	0.858	0.891	0.969	0.741	0.669	0.852
Underhang bearing	0.998	0.992	1.000	0.997	0.960	0.999
Overhang bearing	0.981	0.974	0.995	0.939	0.873	0.955
Pooled	0.956	0.954	0.989	0.897	0.835	0.937

Table 1 is the main result. Conditioning is the best detector for every fault type in both regimes. The invariance baseline is the more revealing comparison: order tracking sometimes edges the unconditional baseline at matched speed, but under speed shift it is consistently *worse* than doing nothing (pooled 0.835 versus 0.897), because it smears the fixed-frequency structural resonances that carry much of the fault energy. Under speed shift, removing speed costs 6.2 AUC points relative to the unconditional detector, while conditioning on the same tachometer signal gains 4.0. The bearing faults are near-saturated at matched speed, so the substantial matched-speed gains concentrate on the harder misalignment and imbalance faults (for example 0.858 → 0.969 on imbalance); the held-out-speed gains are universal.

Across ten random matched-speed splits the pooled figures are baseline 0.956 ± 0.004 (partial AUC at 10% false-positive rate 0.866), invariance 0.954 ± 0.007 (pAUC 0.832), and conditioning 0.989 ± 0.002 (pAUC 0.949), with the conditioning bootstrap interval [0.981, 0.993] clearing the baseline interval [0.941, 0.965]. A paired Wilcoxon test rejects equality of conditioning against both baseline and invariance in the matched regime ($p = 0.005$) and across the six held-out bands ($p = 0.046$ versus baseline, $p = 0.028$ versus invariance).

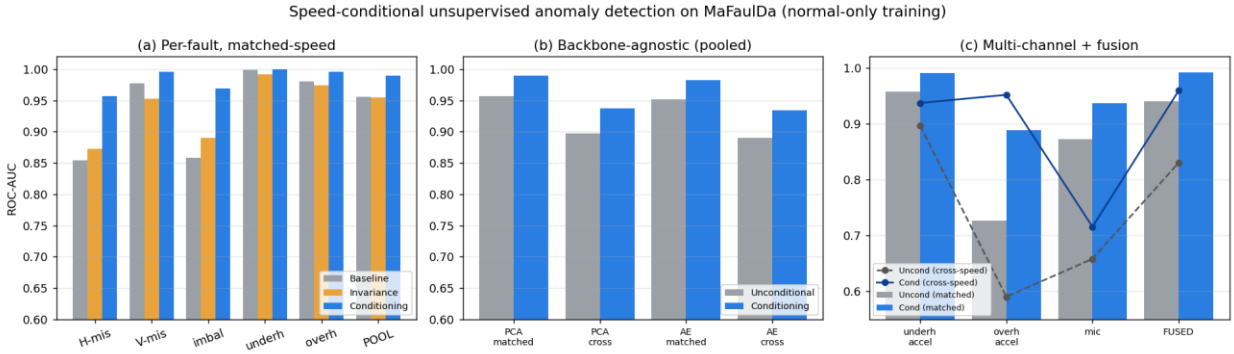


Figure 1. *(a) Per-fault matched-speed AUC for baseline, invariance and conditioning. (b) The conditioning gain is backbone-agnostic: nearly identical under a PCA and an autoencoder detector, and larger than the difference between the backbones. (c) Conditioning wins on every channel including the microphone; multi-channel fusion is below the best single channel without conditioning but best of all with it.*

Backbone-agnostic, and invariance stays behind. Table 2 repeats the pooled comparison under two backbones, a PCA reconstruction and a dense autoencoder, and adds the invariance arm at each. Three things hold. The conditioning gain is essentially unchanged across backbones (+0.033/+0.040 for PCA, +0.032/+0.043 for the autoencoder), and the autoencoder alone does not beat PCA alone, so the improvement comes from conditioning, not from backbone capacity. The invariance approach does not recover when given the stronger backbone: order tracking under the autoencoder still collapses under speed shift (0.809), below the unconditional detector. Nor is this a matter of under-tuning the invariance arm. Sweeping its detector, PCA rank from 16 to 256 and autoencoder latent width from 16 to 64, moves its held-out-speed AUC only within [0.78, 0.837]; the best configuration, 0.837, still sits below the unconditional baseline (0.897), because the limitation is representational, order tracking discards

the fixed-frequency fault energy, not a matter of detector capacity. Since the strongest recent speed-invariance method for this setting, OASSL [9], is not publicly available, we take an order-tracked spectrum with a tuned reconstruction detector as its reproducible in-class representative; conditioning outperforms it at both backbones and across the tuning sweep.

Table 2. *Pooled ROC-AUC (matched / held-out speed) by backbone and treatment. The conditioning gain is backbone-agnostic, and the invariance (order-tracking) arm stays below the unconditional baseline under speed shift at both backbones.*

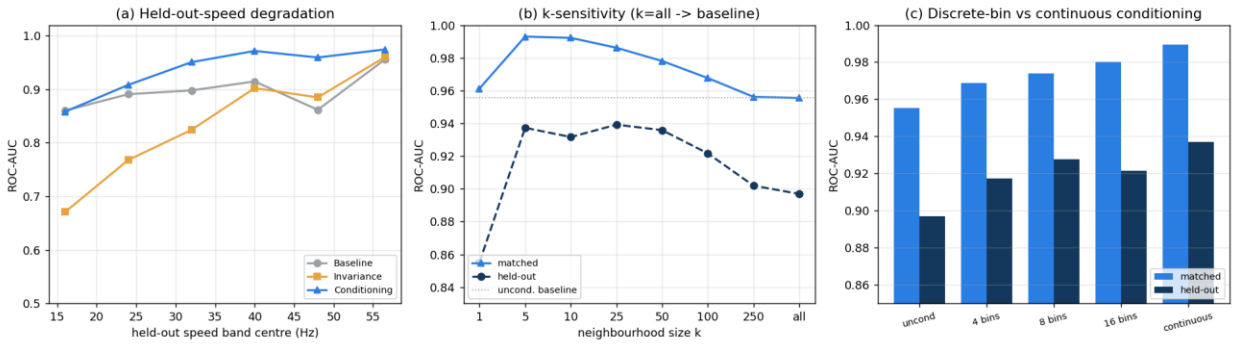
Backbone	Unconditional	Invariance	Conditioning
PCA	0.956 / 0.897	0.954 / 0.835	0.989 / 0.937
Autoencoder	0.952 / 0.890	0.960 / 0.809	0.984 / 0.933

Across channels and fusion. Table 3 varies the sensor channel. Conditioning wins on both accelerometers and on the microphone. Two points stand out. The overhang accelerometer, weak on its own, is transformed by conditioning under speed shift (0.591 → 0.952). And fusion of the three channels, by averaging standardized scores, is *below* the best single channel without conditioning (0.940 < 0.957, because the weak channels drag the average down) but the best detector of all *with* conditioning (0.992/0.959): conditioning equalizes the channels and is what makes fusion pay.

Table 3. *Pooled ROC-AUC by channel (matched / held-out speed). Conditioning wins on every channel; fusion helps only with conditioning.*

Channel	Unconditional	Conditioning
Underhang accelerometer	0.957 / 0.897	0.990 / 0.937
Overhang accelerometer	0.726 / 0.591	0.889 / 0.952
Microphone	0.873 / 0.658	0.936 / 0.716
Fusion (all three)	0.940 / 0.830	0.992 / 0.959

Ablations: neighbourhood size and continuous versus discrete conditioning. Figure 2 examines the method's single hyperparameter and its central design choice. The neighbourhood size k has a broad optimum: matched-speed AUC peaks near $k = 5$ and held-out AUC is flat over $k \in [5,50]$ (panel b). At the degenerate limit $k = n$ the conditional mean becomes the global mean and the detector reproduces the unconditional baseline to three decimals (0.956 / 0.897), a built-in consistency check that the implementation passes. Panel (c) and Table 4 answer the question the novelty claim raises: is a continuous covariate necessary, or does any conditioning suffice? Binning the speed into 4, 8, or 16 bins and subtracting the per-bin mean already improves over the unconditional baseline (0.955 to 0.980 matched as the bins refine), which shows conditioning helps at all; continuous nearest-neighbour conditioning then improves further to 0.989, above the finest binning, which shows the continuous covariate carries value a discrete domain label does not.



*kk = n**Figure 2.** (a) Held-out-speed degradation: conditioning holds above the baseline at every held-out band while invariance falls below it, most at low speed. (b) Sensitivity to the neighbourhood size ; the dotted line is the unconditional baseline, which ("all") recovers. (c) Continuous nearest-neighbour conditioning exceeds discrete-bin conditioning at every bin count.*

***Table 4.** Continuous versus discrete-bin conditioning, pooled ROC-AUC (matched / held-out speed). Conditioning on the continuous covariate exceeds conditioning on a binned version of it, the form the nearest prior work takes.*

Conditioning	Matched	Held-out
None (unconditional)	0.955	0.897
Discrete, 4 bins	0.969	0.917
Discrete, 8 bins	0.974	0.928
Discrete, 16 bins	0.980	0.922
Continuous (nearest-neighbour)	0.989	0.937

Mechanism control. A covariate-shuffle control establishes that the gain requires the true speed labels, not merely the detrending operation. On the primary configuration, detrending healthy-test windows with their true speeds gives AUC 0.961; detrending the same windows with randomly permuted speed labels collapses it to 0.419, with the mean healthy score rising accordingly. The normal-versus-normal null sits at 0.50 for all methods, confirming the detectors do not key on the train/test split.

A second dataset, and the scope of the operating covariate. The principle is not specific to MaFaulDa. On MIMII-DG [5], the DCASE 2022 domain-generalization benchmark of machine sounds, each machine provides an operating attribute and a source/target domain split; we condition a log-mel reconstruction detector on that attribute and evaluate all five machine types (Table 5). Conditioning improves both domains on the four machines whose attribute is a genuine operating covariate: bearing rotation speed, gearbox voltage, slider velocity, and the valve operation pattern. The gain is sharpest on the shifted target domain: valve target detection, which the unconditional detector drives below chance (0.376) because the domain shift makes target-normal sound anomalous, is restored to 0.632 by conditioning on the operation pattern. The exception is the fan, whose domain attribute is a categorical mixture of background machine noises rather than a machine operating state; there conditioning gives no benefit, which is the expected boundary of the method. The pattern across machines is therefore consistent with the thesis: conditioning helps when the covariate is a genuine operating condition, on both continuous covariates (speed, voltage, velocity) and an ordinal discrete one (valve pattern). These per-machine source and target AUCs isolate the conditioning effect on a deliberately simple detector; they are not a challenge submission, whose official score aggregates AUC and partial AUC across all machines by harmonic mean.

Table 5. *Cross-dataset breadth on MIMII-DG (DCASE 2022 domain generalization), ROC-AUC per machine and domain. Conditioning helps on both domains for the four machines with a genuine operating covariate; the fan, whose attribute is a categorical background-noise mixture, is the expected exception.*

Machine (attribute)	Domain	Unconditional	Conditioning
Bearing (rotation speed)	source	0.609	0.649
Bearing (rotation speed)	target	0.623	0.645
Gearbox (voltage)	source	0.748	0.809
Gearbox (voltage)	target	0.645	0.710
Slider (velocity)	source	0.726	0.789
Slider (velocity)	target	0.622	0.662
Valve (operation pattern)	source	0.690	0.734
Valve (operation pattern)	target	0.376	0.632
Fan (noise mixture)	source	0.955	0.943
Fan (noise mixture)	target	0.716	0.670

7 Discussion and limitations

The finding is a design lesson: when a normal-only detector must operate across a measured operating condition, conditioning on that covariate beats both ignoring it and removing it. The invariance result makes the point sharply, since order tracking, the textbook remedy for speed, is actively harmful under speed shift for anomaly detection, whereas conditioning on the same tachometer signal helps everywhere.

When conditioning helps, and when it cannot. Conditioning interpolates the healthy manifold across the operating range; it does not extrapolate beyond the speeds present in training. When the healthy training data spans the operating range, as in the leave-one-band-out protocol where every held-out band is bracketed by training speeds, the local mean at an unseen speed is a reliable interpolant and detection improves. When training covers only a narrow band, the nearest-neighbour mean for a distant query is a mismatched interpolant and conditioning offers no advantage there. This bounds the method precisely: it needs healthy data across the operating envelope, which is the realistic condition for a machine monitored over its duty cycle. The effect is visible in the per-band results, where the weakest conditioning gains sit at the slowest band, the edge of the training support. We also note that the held-out-speed AUC is a per-band average, each band scored independently, and is therefore a stricter per-operating-point measure than the pooled matched-speed AUC rather than a directly comparable number.

The study is validated on the MaFaulDa test rig, where the operating condition is a single clean scalar (rotation speed) with a reliable tachometer, with the MIMII-DG results extending it to an acoustic benchmark and a discrete covariate. Extending the conditioning to additional continuous covariates (load, temperature) and to settings where the covariate is noisier or must be estimated from the signal is future work. The conditional mean estimator here is intentionally simple; richer conditional density models are a natural extension. Within the studied setting the effect is consistent across fault types, backbones, channels, a second dataset, and a causal control.

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11. **Data and code.** MaFaulDa (SMT/COPPE/UFRJ) and MIMII-DG (Zenodo 10.5281/zenodo.6529888) are public. Code, the experiment registry, and every result artifact are at github.com/ApartsinProjects/EngineKnock; all headline numbers derive from a single consolidated evaluation artifact.

Draft · results reproducible from the EngineKnock experiment registry.